Build Your First AI Chatbot Using Python

Summary:-

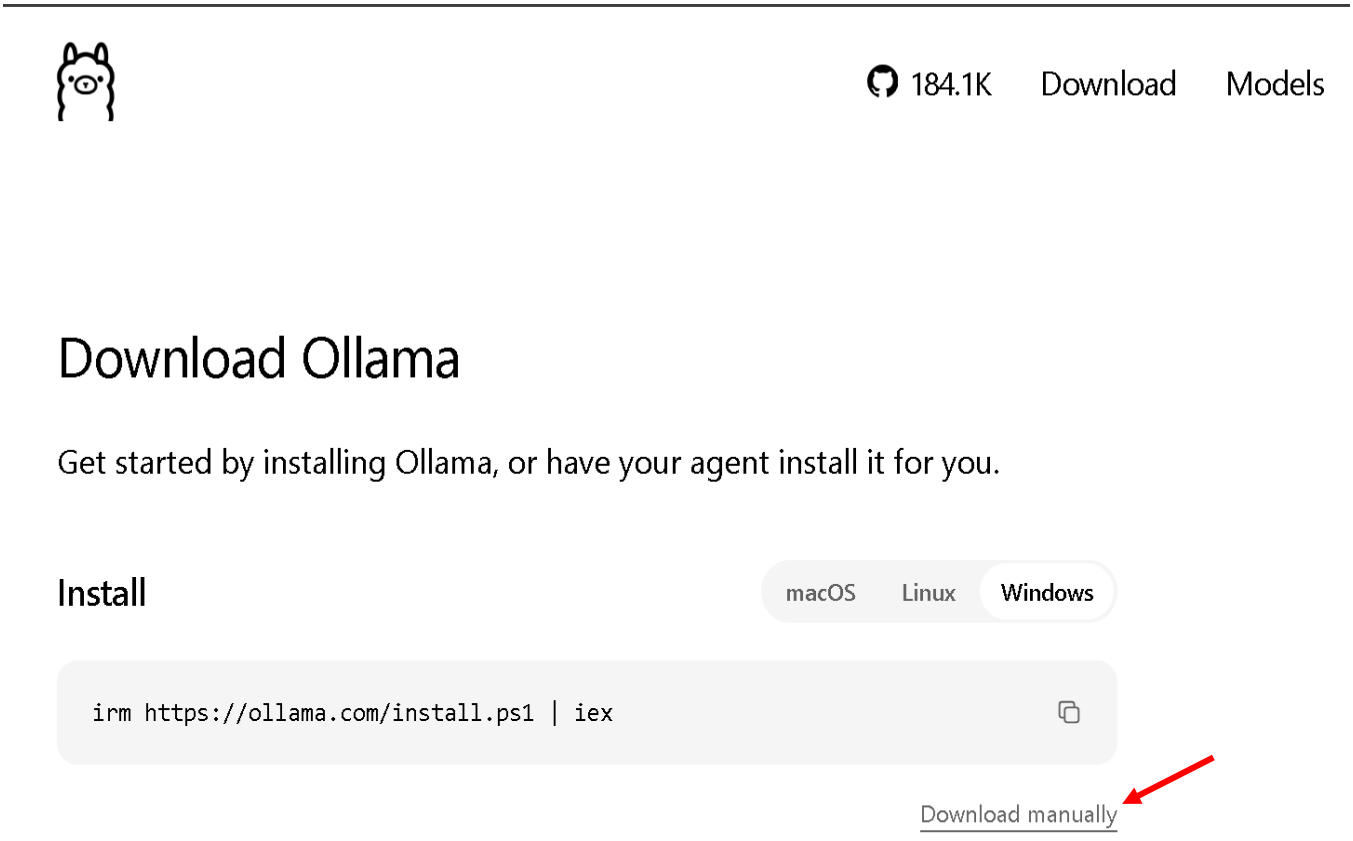
We created a basic AI chatbot using Python, Ollama, and the Qwen3:4B model. Using VS Code and the OpenAI Python library, we developed an interactive assistant that answers user questions through the terminal without requiring a paid cloud AI API.

To Build AI Applications we'll use free tools: -

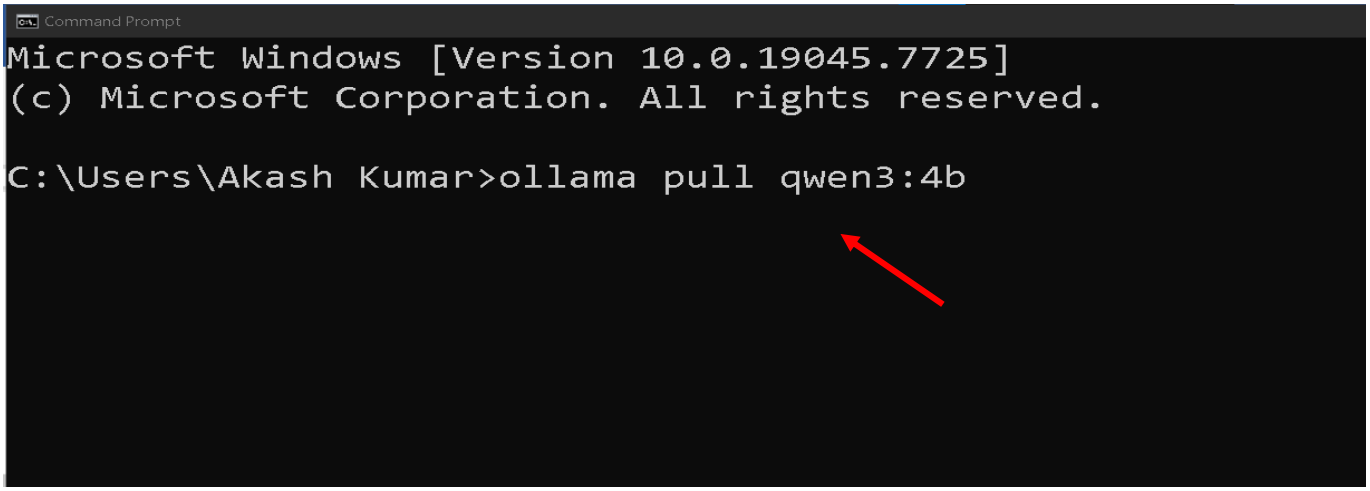
- Python
- Visual Studio Code
- Ollama
- Git

Lets Start: -

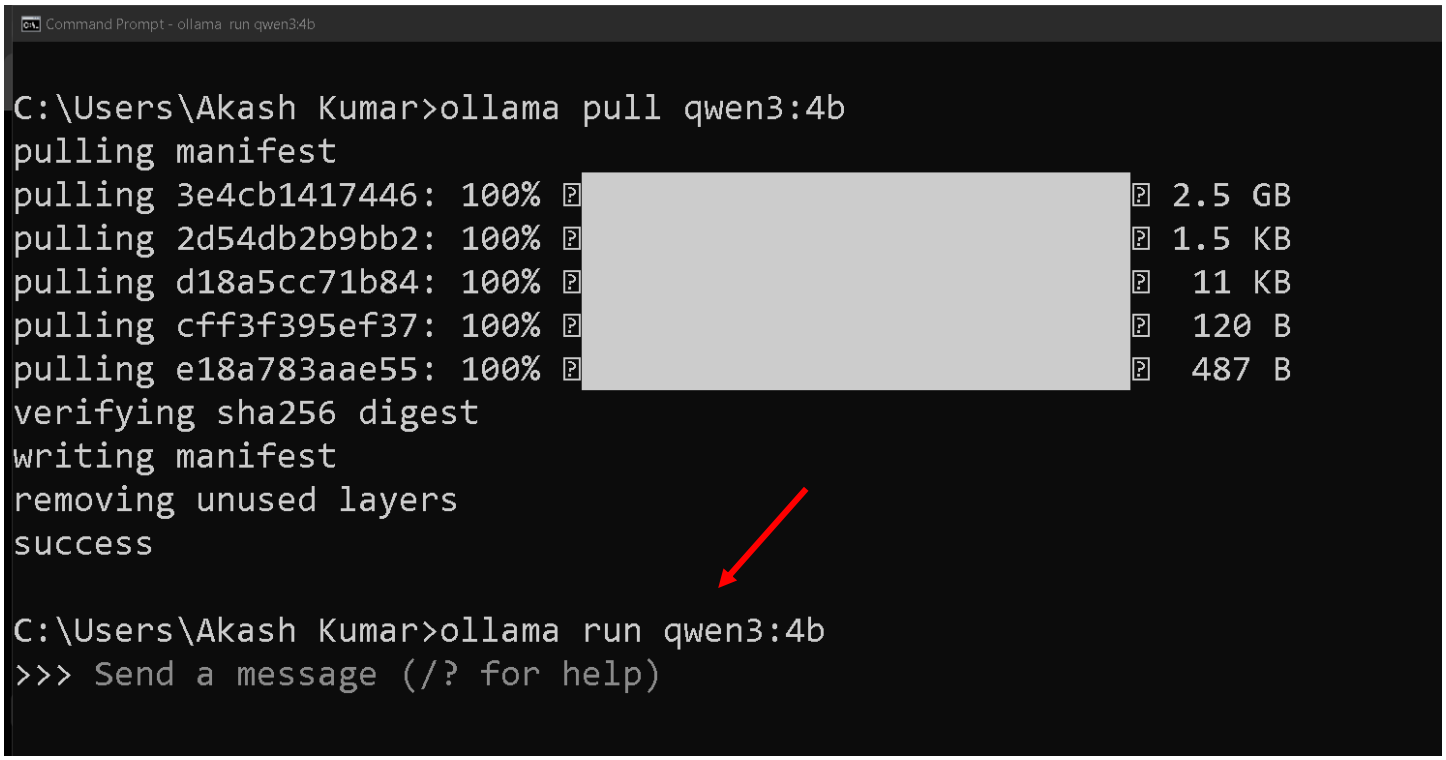
- First download Ollama - Link



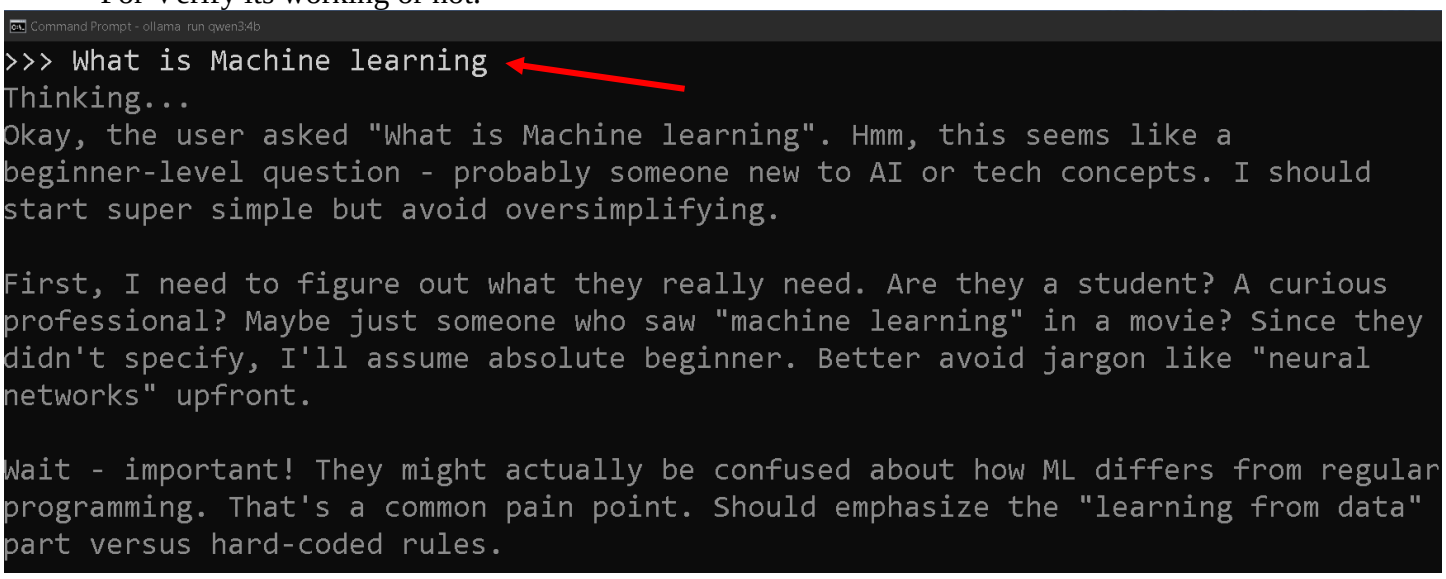
- After downloading Install Ollama.
- Then After installation open command prompt.



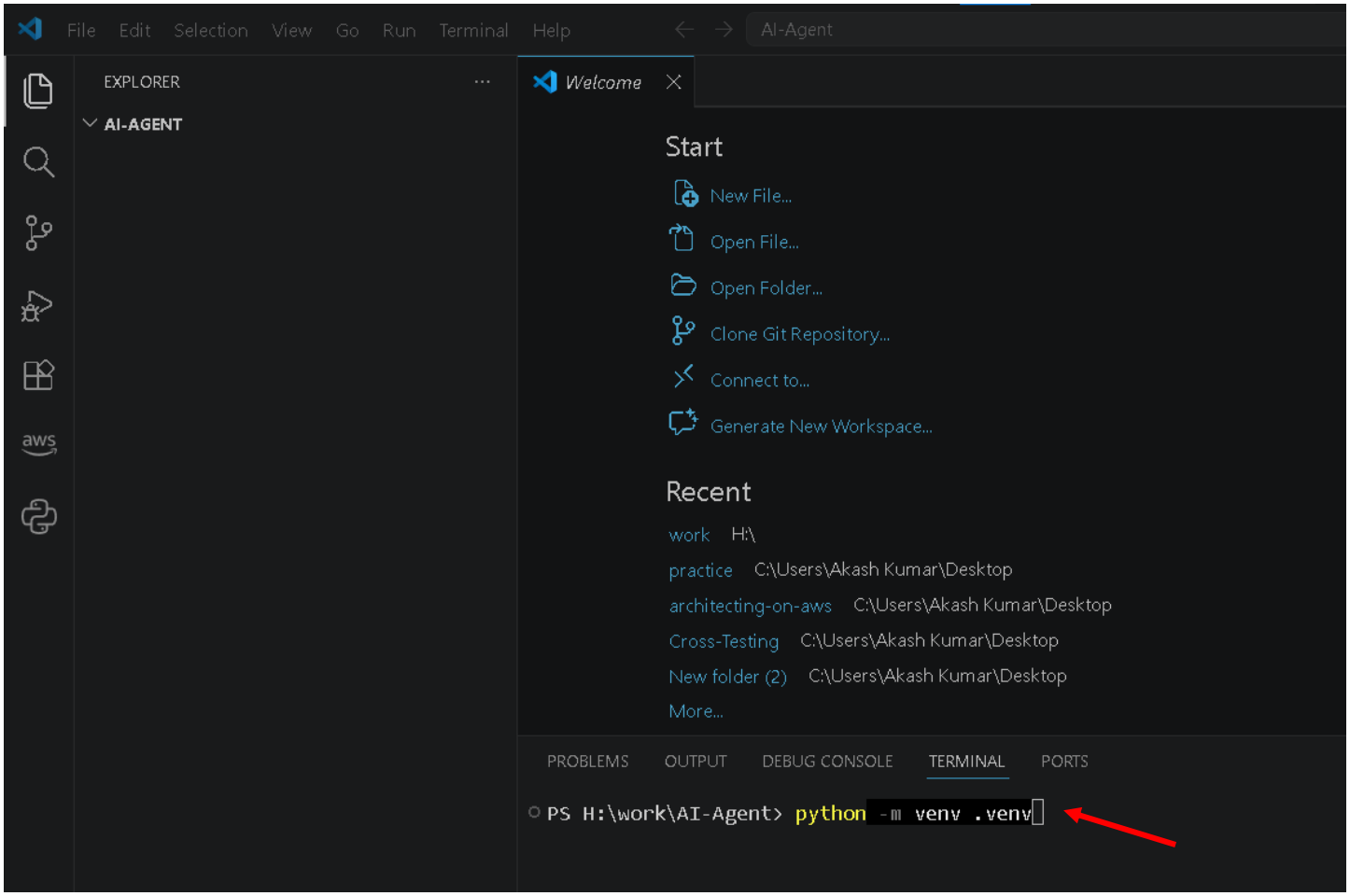
- And Enter ollama pull qwen3:4b
- It will take time.



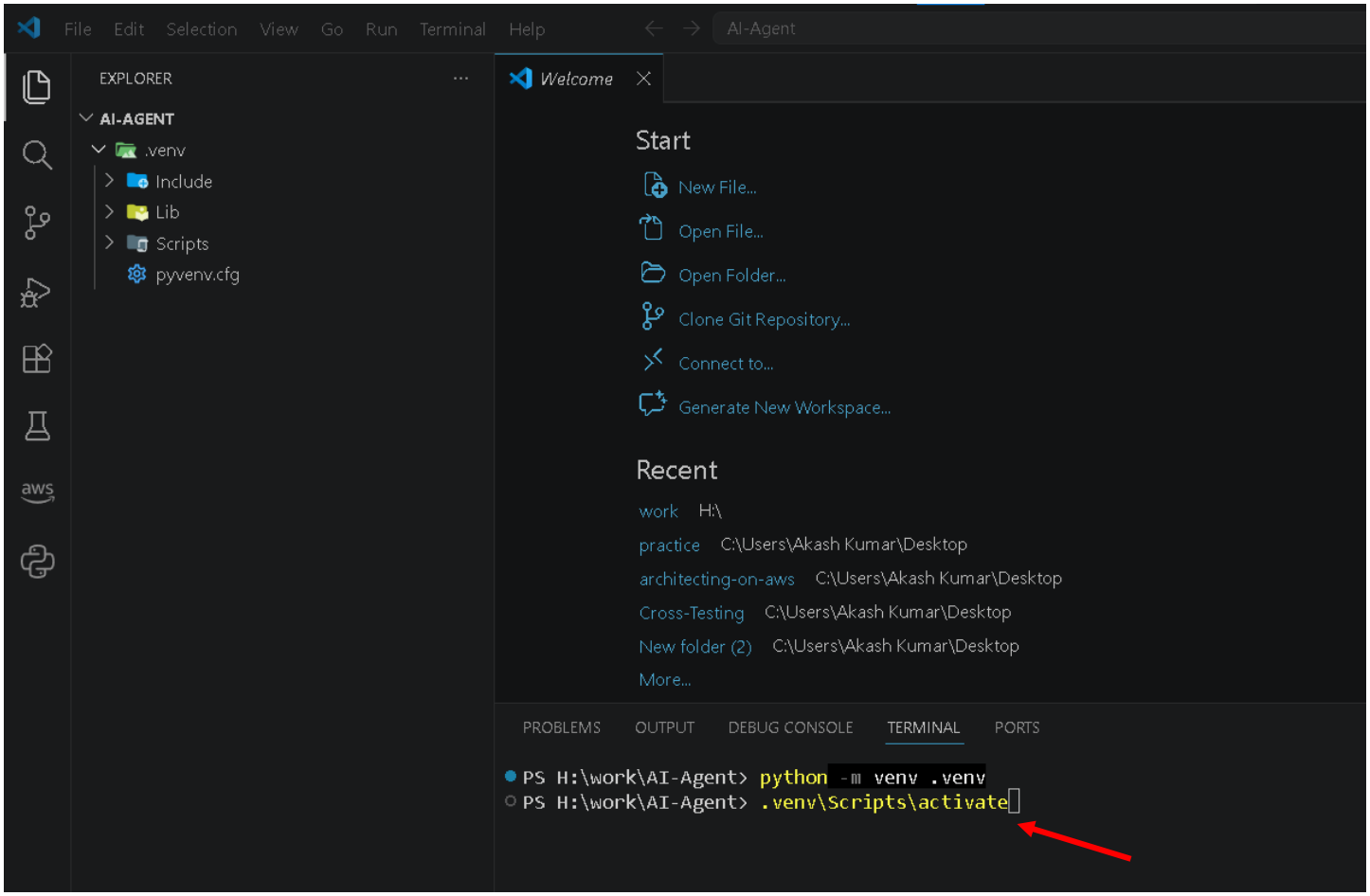
- Now after Downloading AI model.
- Enter ollama run qwen3:4b
- For Verify its working or not.



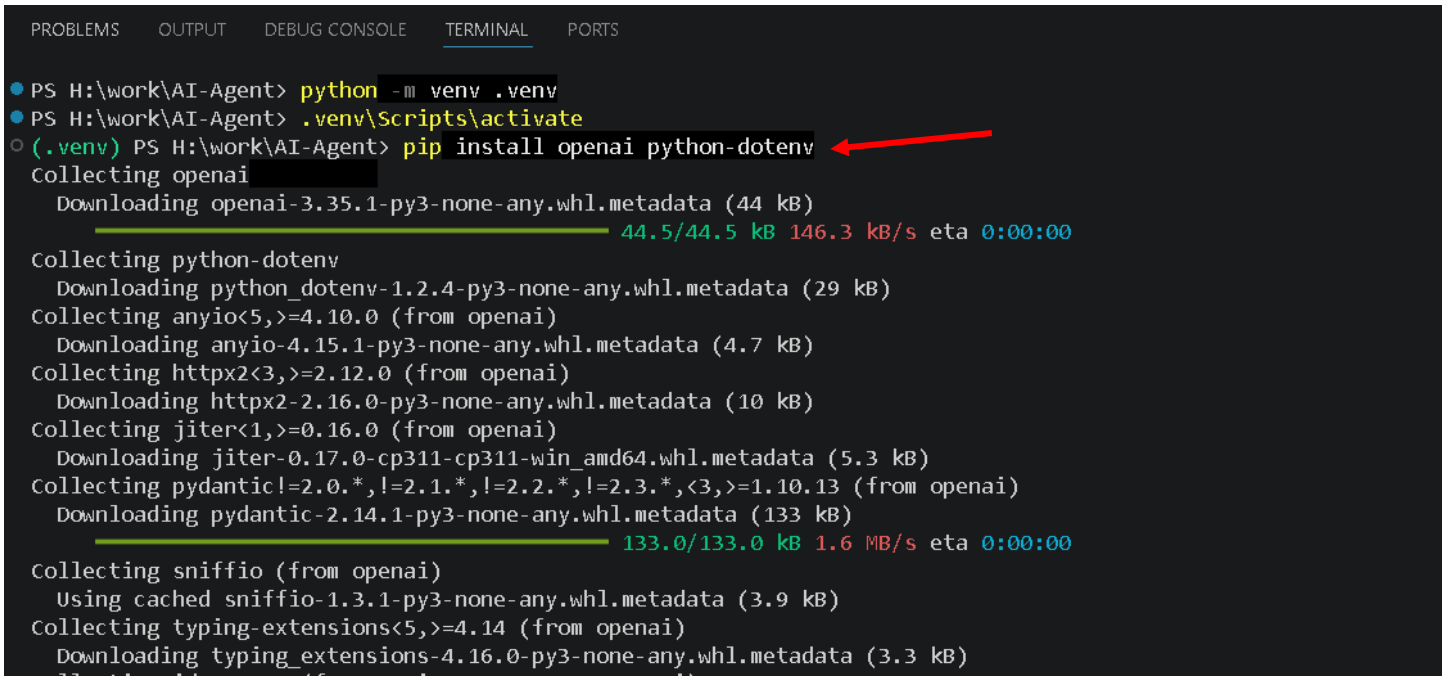
- ask some questions and check its work properly.
- Now create your project folder in your system and open in Vs Code.



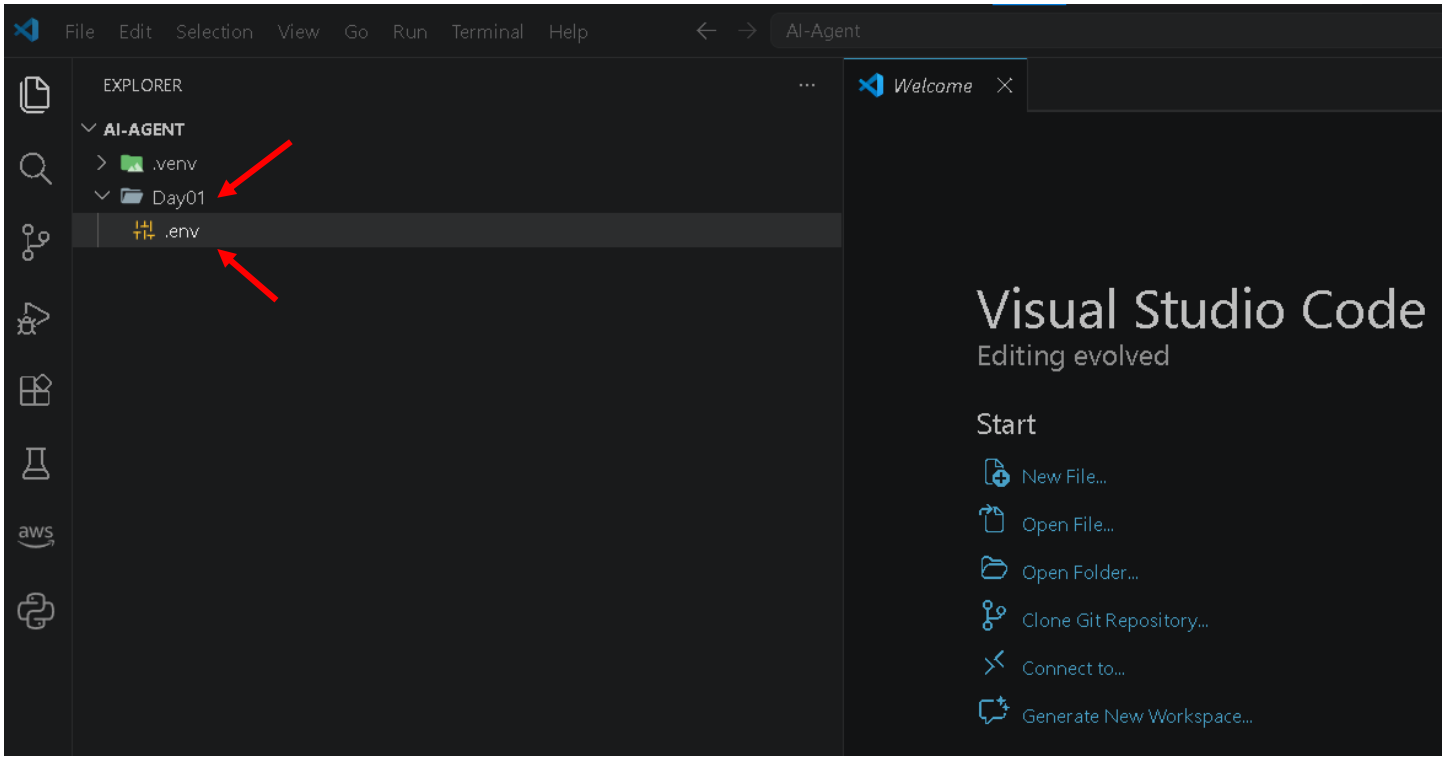
- Our Folder AI-AGENT is opened in Vs Code.
- Now we need to create virtual environment .
- So open terminal and Enter python -m venv .venv



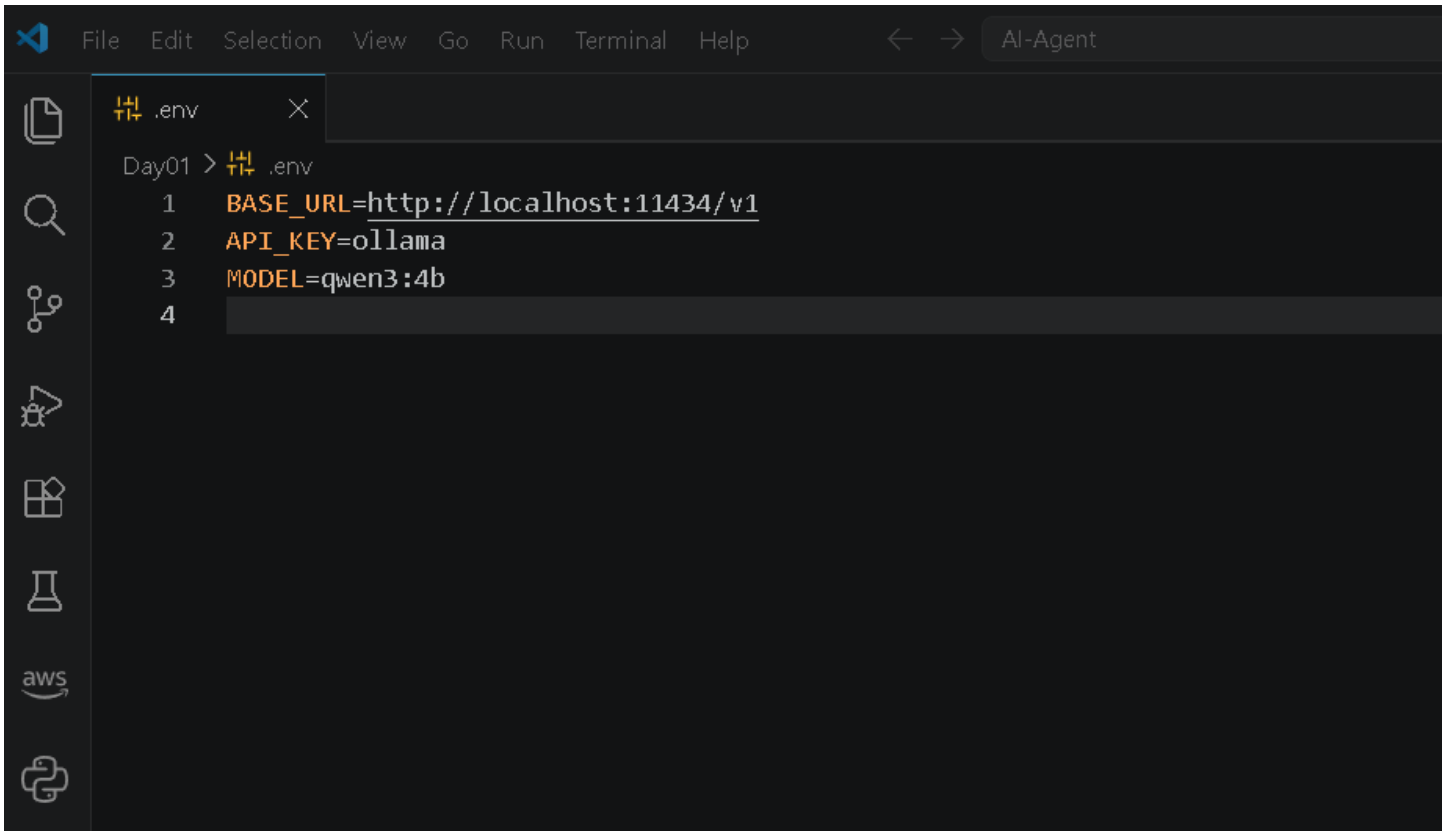
- Now when virtual environment is created we need to activate,
- So enter .venv\Scripts\activate
- Now we need to install some python Library.



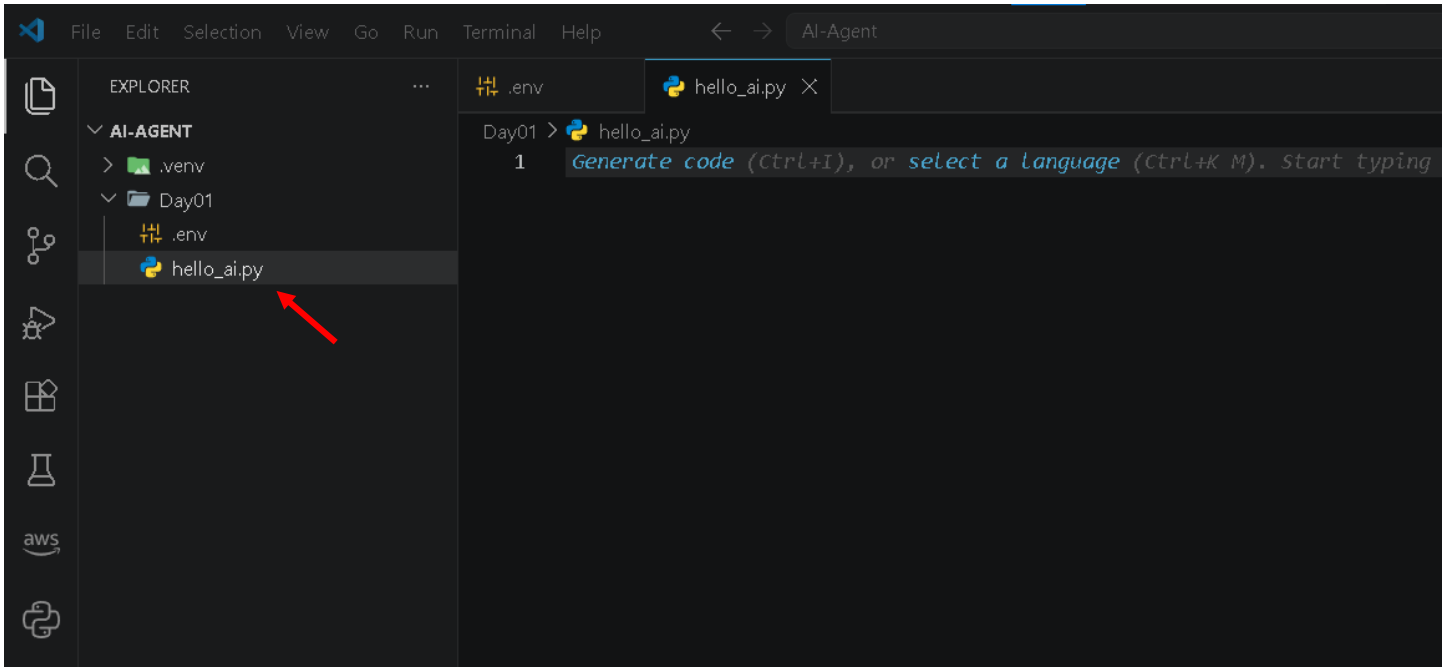
- So We Enter pip install openai python-dotenv



- Now After installation create folder inside AI-AGENT.
- And inside new folder create a new file .env



- Now open .env file and Enter these things:-
- BASE_URL=http://localhost:11434/v1
- API_KEY=ollama
- MODEL=qwen3:4b



- Now create New file in same folder hello_ai.py
- And Enter the Following codes: -

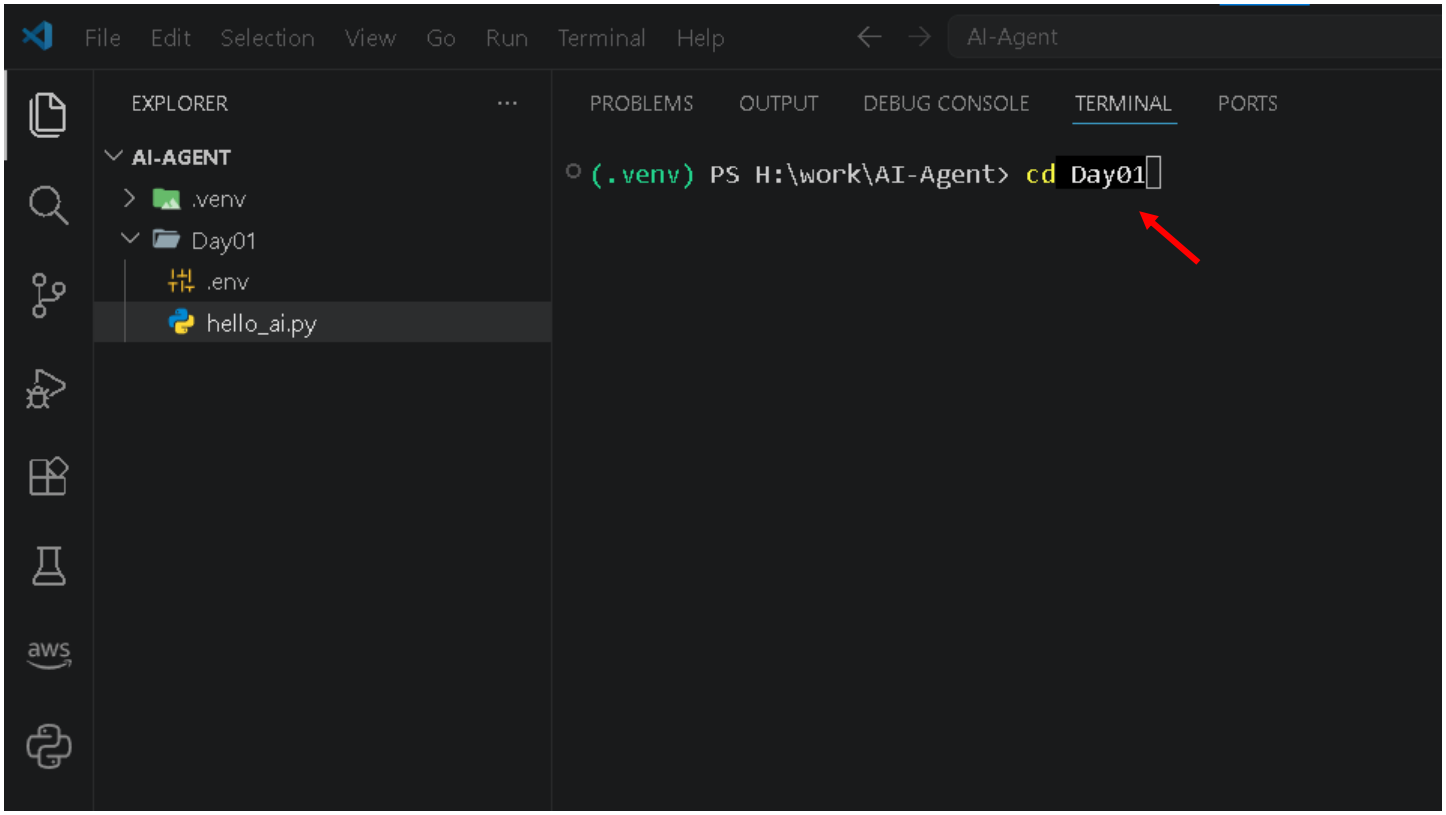
```
from openai import OpenAI
from dotenv import load_dotenv
import os

# Load configuration from the .env file
load_dotenv()

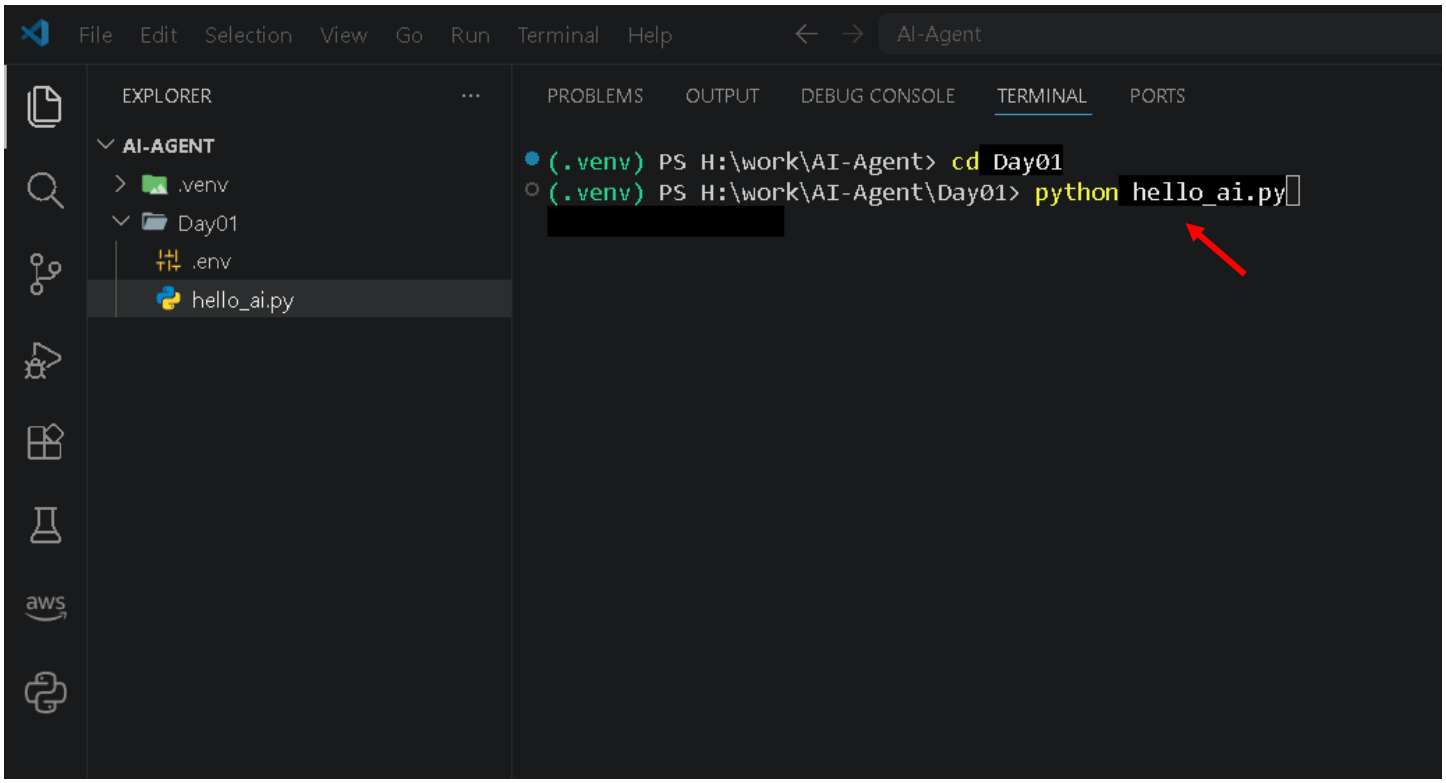
# Create a client that communicates with Ollama
client = OpenAI(
    base_url=os.getenv("BASE_URL"),
    api_key=os.getenv("API_KEY")
)

# Send a question to the AI model
response = client.chat.completions.create(
    model=os.getenv("MODEL"),
    messages=[
        {
            "role": "user",
            "content": "What is Artificial Intelligence?"
        }
    ]
)

# Display the response
print(response.choices[0].message.content)
```



- Now open the terminal and go inside Day01 directory.
- So for this enter cd Day01 in terminal.



- Now Enter python hello_ai.py

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS                                   powershell - Day01

(.venv) PS H:\work\AI-Agent\Day01> python hello_ai.py
Artificial Intelligence (AI) is a **branch of computer science that focuses on creating systems capable of performing tasks that typically require huma
n intelligence**. These systems learn, reason, solve problems, perceive, understand language, and make decisions—often without explicit programming for
 *each specific task*.

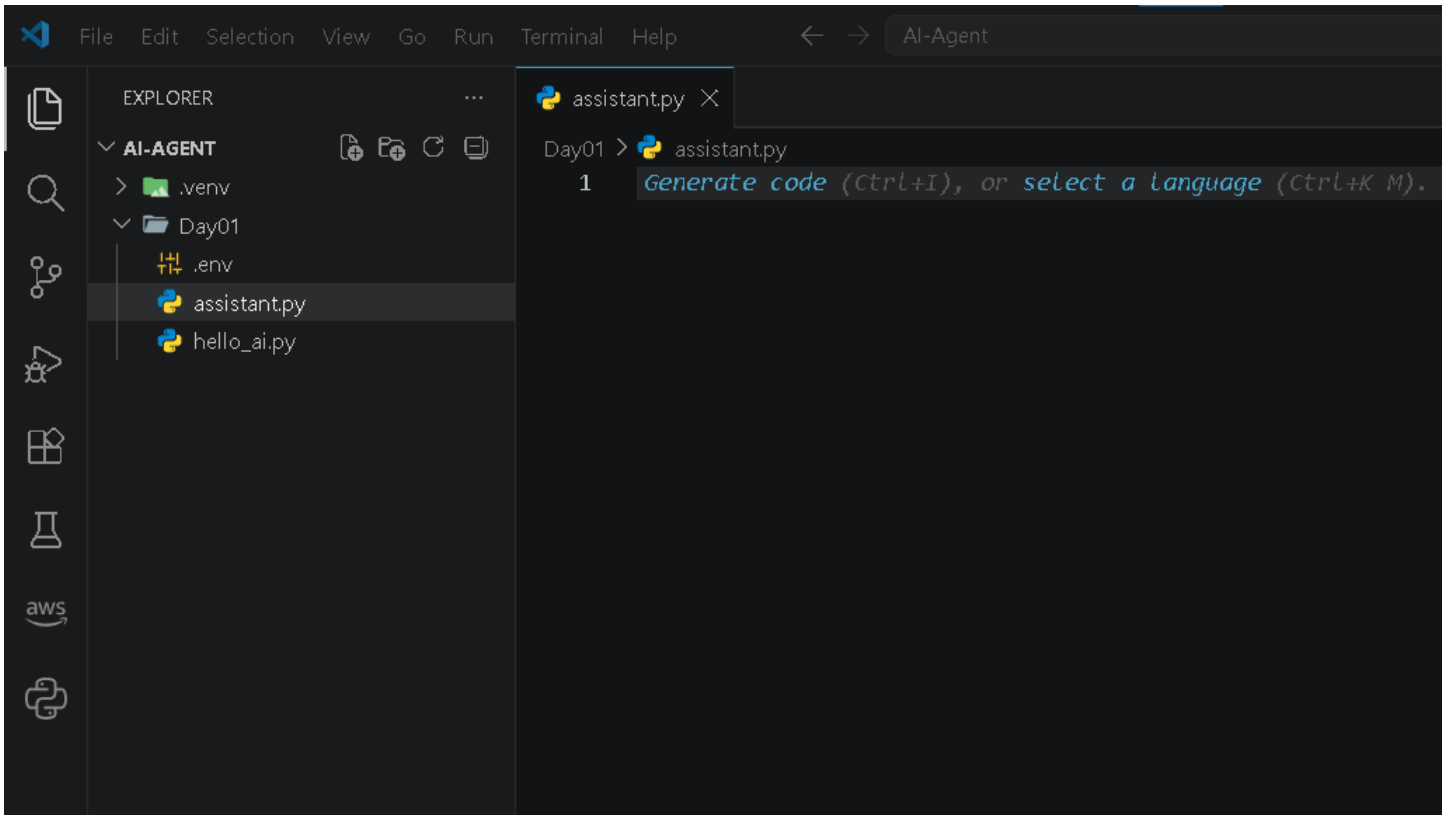
### ★ Core Idea in Simple Terms:
**AI is technology that lets machines "think" and act like humans** (or simulate human-like intelligence) to accomplish complex tasks. It's *not* about
 machines "being smart" like humans—but about **programming computers to mimic intelligence** in ways useful for real-world problems.

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### 🔧 Key Characteristics of AI (What Makes It "Intelligent" in This Context):
| Feature              | What It Means                                                                 | Real-World Example                    |
|----------------------|-------------------------------------------------------------------------------|---------------------------------------|
| **Learning**         | Systems improve at tasks through data/experience (e.g., pattern recognition). | Netflix recommendations, spam filters |
| **Problem Solving**  | Finding solutions to complex, unstructured problems.                          | Google Maps finding the fastest route |
| **Perception**       | Interpreting sensory input (e.g., vision, sound).                             | Self-driving cars identifying pedestrians |
| **Language**         | Understanding/communicating in human language.                                | Chatbots like ChatGPT, virtual assistants |
| **Decision-Making**  | Choosing actions based on data, goals, and risk.                              | Financial trading algorithms          |

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```

- Now see it will return output according our given content in our code.
- But one issue is that if we need to chat about new topic we need to go inside code and change content.
- So now we can resolve it.



- Now we create a new python file assistant.py
- Now enter the following code: -

```
from openai import OpenAI
from dotenv import load_dotenv
import os

# Load configuration
load_dotenv()

# Create AI client
client = OpenAI(
    base_url=os.getenv("BASE_URL"),
    api_key=os.getenv("API_KEY")
)
```

```
print("=" * 40)
print("      My AI Assistant")
print("=" * 40)

while True:

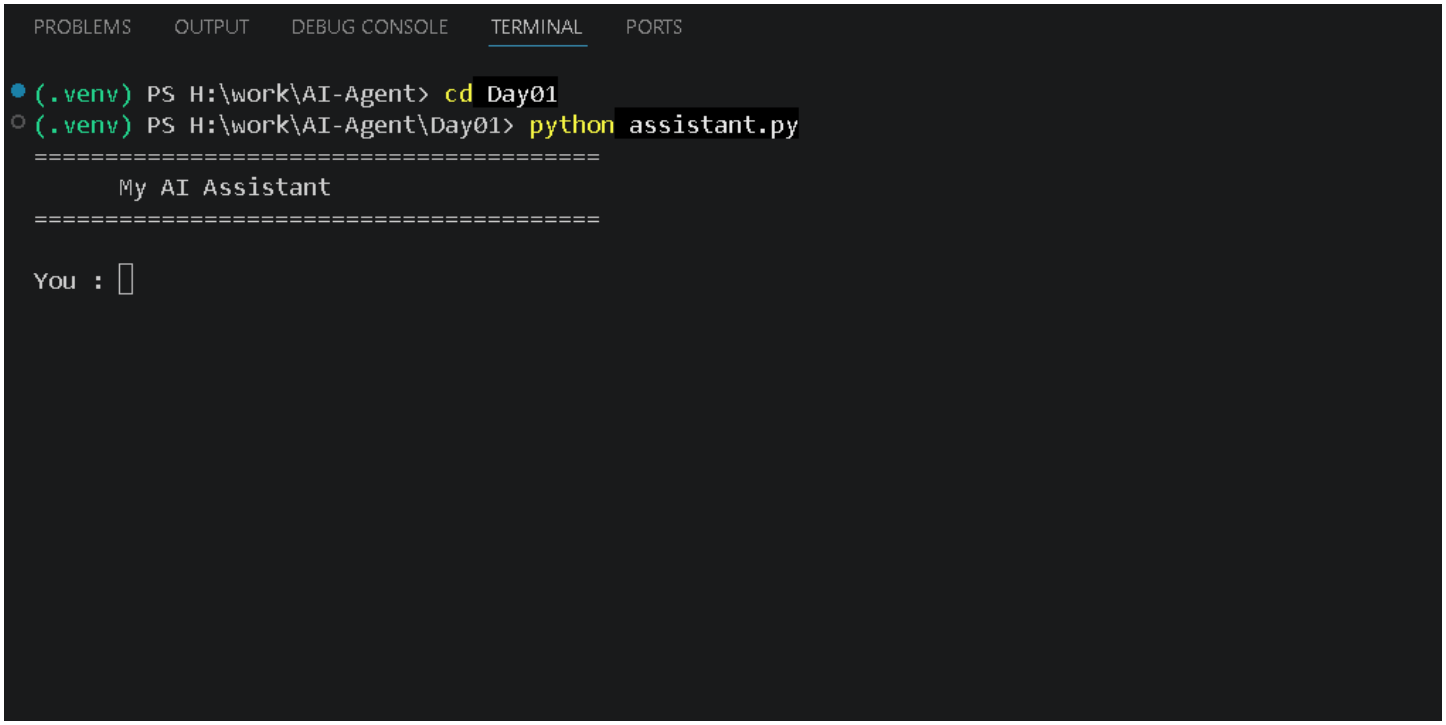
    user_input = input("\nYou : ")

    if user_input.lower() == "quit":
        print("\nAI  : Goodbye! Have a great day.")
        break

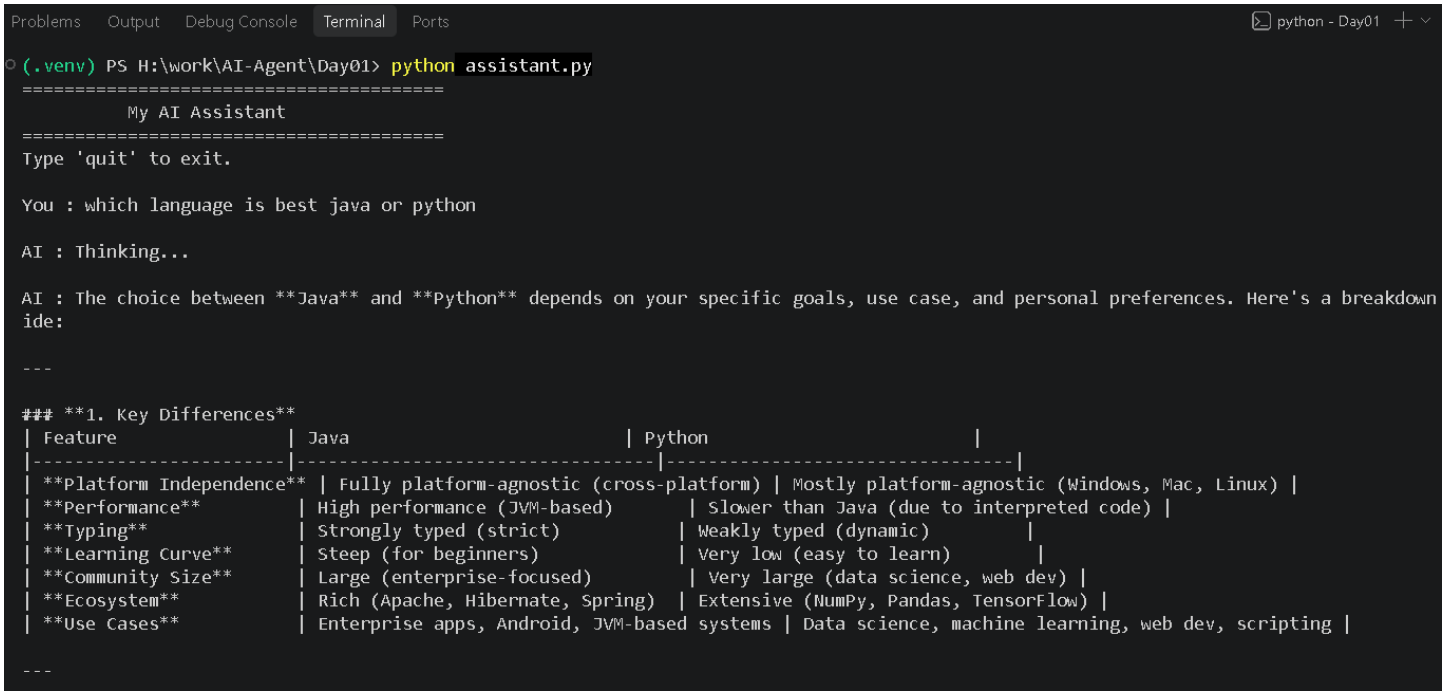
    response = client.chat.completions.create(
        model=os.getenv("MODEL"),
        messages=[
            {
                "role": "user",
                "content": user_input
            }
        ]
    )

    print("\nAI :", response.choices[0].message.content)
```

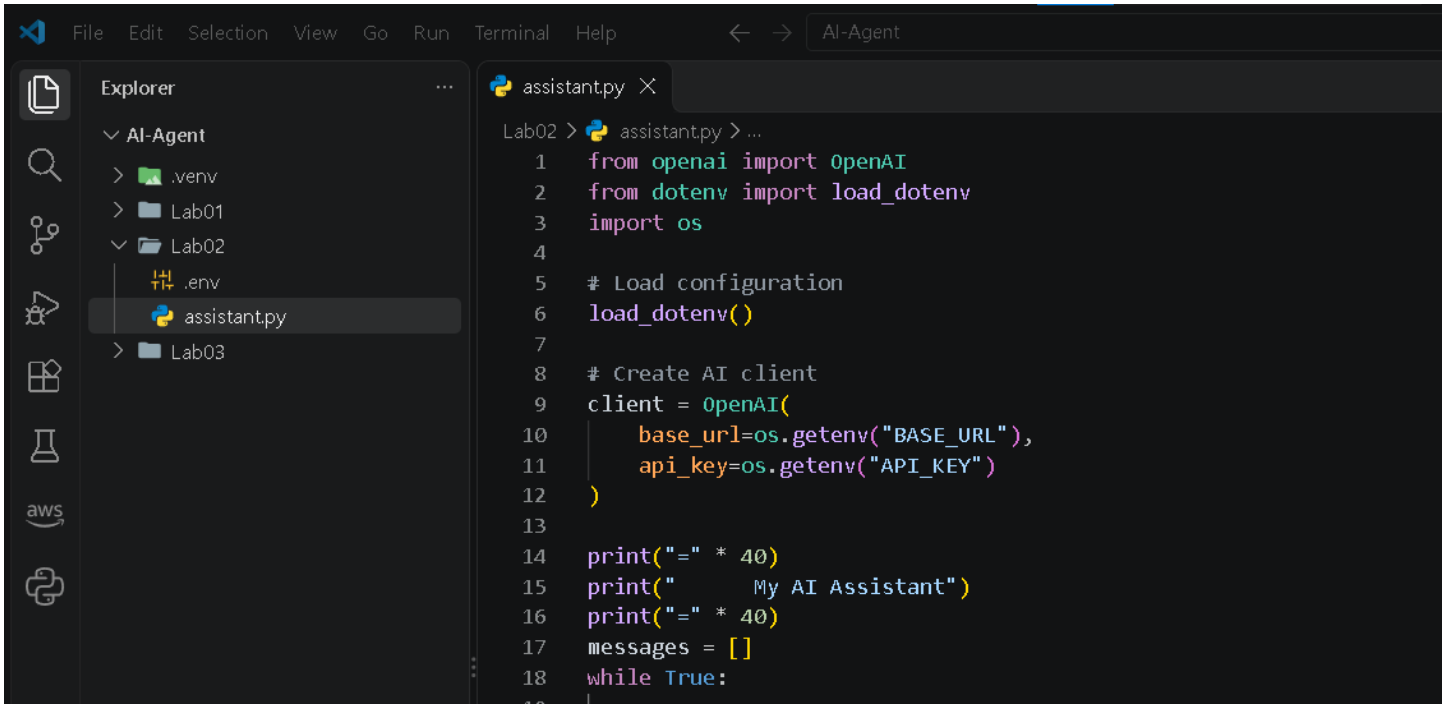
- Now open terminal.



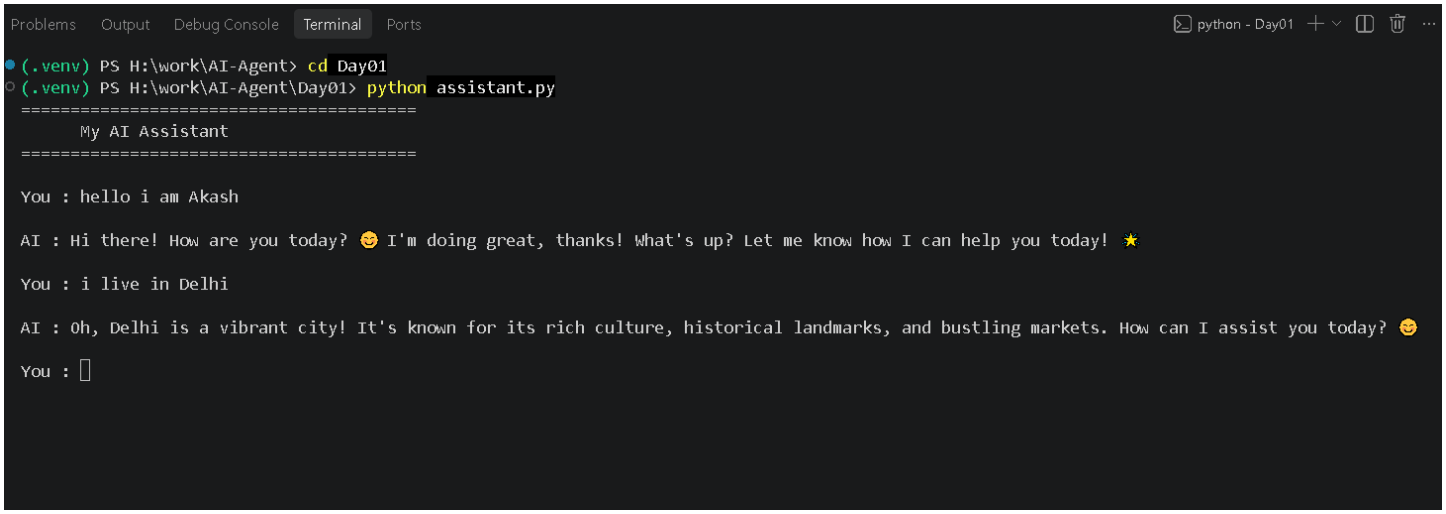
- And Enter python assistant.py
- Now see our assistant ask topic in our terminal not in code.
- Now check its work perfectly.



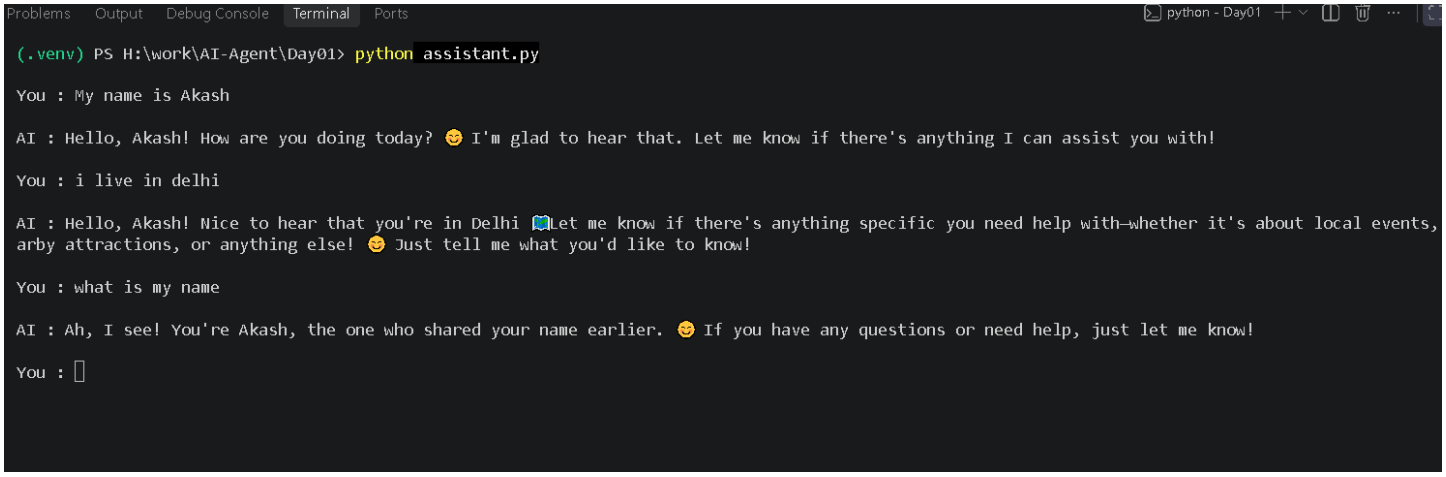
- Now you see I ask which language is best java or python.
- Out chat bot reply perfectly.
- Now we need to provide memory to our assistant.



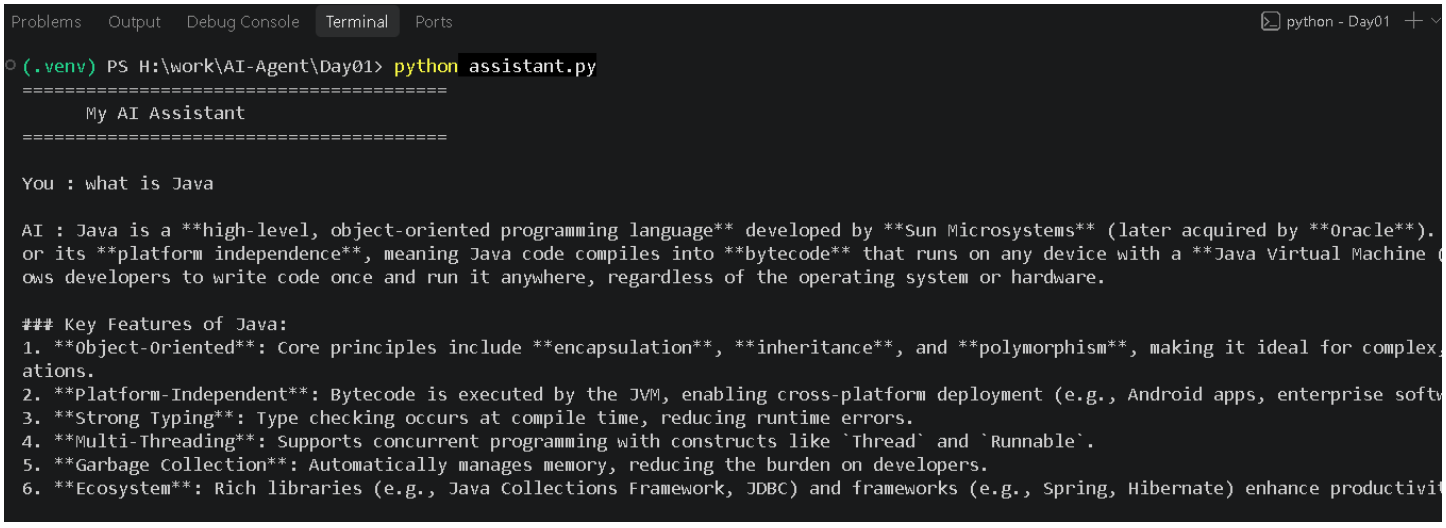
- Now we create a folder Lab02 where we work on Memory part.
- So we update our code assistant.py
- Code is provided in resource.



- After updating my assistant.py code
- I give some input about me.
- Now let's see it work correctly and give output.



- Now see I ask what is my name it tells my name.
- Now we need to add a conversation memory.
- It means our agent remember our previous chats.
- So we update our assistant.py code.
- Updated code is provided in resource.



- Now see I ask what is java then it provides output.
- And when you scroll down you see conversation history.

```
Problems  Output  Debug Console  Terminal  Ports                                   python - Day01

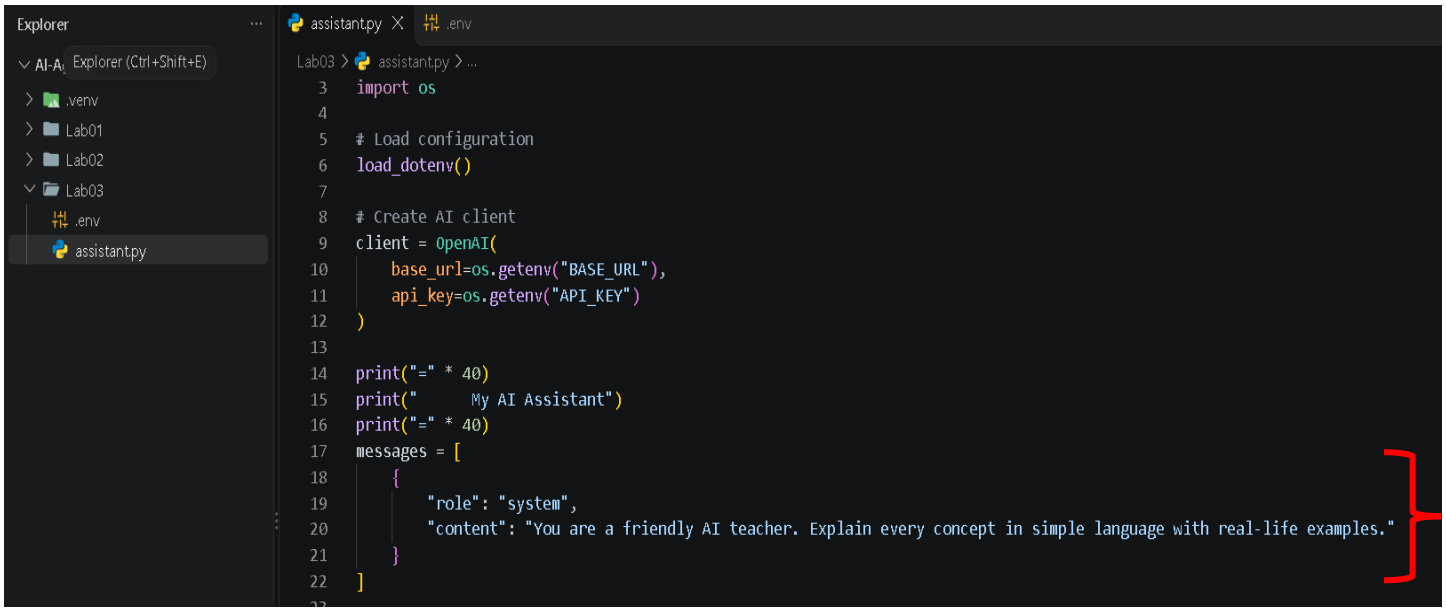
(.venv) PS H:\work\AI-Agent\Day01> python assistant.py

========== Conversation History ==========
User : what is Java
Assistant : Java is a **high-level, object-oriented programming language** developed by **Sun Microsystems** (later acquired by **Oracle**). It is ren
owned for its **platform independence**, meaning Java code compiles into **bytecode** that runs on any device with a **Java Virtual Machine (JVM)**. T
his allows developers to write code once and run it anywhere, regardless of the operating system or hardware.

### Key Features of Java:
1. **Object-Oriented**: Core principles include **encapsulation**, **inheritance**, and **polymorphism**, making it ideal for complex, scalable applic
ations.
2. **Platform-Independent**: Bytecode is executed by the JVM, enabling cross-platform deployment (e.g., Android apps, enterprise software).
3. **Strong Typing**: Type checking occurs at compile time, reducing runtime errors.
4. **Multi-Threading**: Supports concurrent programming with constructs like `Thread` and `Runnable`.
5. **Garbage Collection**: Automatically manages memory, reducing the burden on developers.
6. **Ecosystem**: Rich libraries (e.g., Java Collections Framework, JDBC) and frameworks (e.g., Spring, Hibernate) enhance productivity.

### Use Cases:
- **Android Development**: Android apps are primarily built using Java (though Kotlin is now the preferred language).
- **Enterprise Software**: Used for building robust, scalable applications.
- **Web Applications**: Java is used in frameworks like **Java Servlets** and **Java Spring Boot**.
- **Desktop Applications**: Java Swing and JavaFX for graphical user interfaces.
```

- Basically this conversation history is for debugging process.
- So our memory work is done.
- Now we work on our Assistant Personality on Lab03.



- Now we blank message in assistant.py
- And add Message with role (System) and content (what you want that your assistant behave like).
- In my scenario I give content (You are a friendly AI teacher. Explain every concept in simple language with real-life examples)
- Code is provided in resource.
- Now we see how our assistant behave if we give topic.

```
Problems  Output  Debug Console  Terminal  Ports                                                                    python

(.venv) PS H:\work\AI-Agent> cd Lab03
(.venv) PS H:\work\AI-Agent\Lab03> python assistant.py
========================================
      My AI Assistant
========================================

You : Explain loops

AI : Sure! Let me explain loops with a **simple, relatable example**! 😊

---

### What is a Loop?
A **loop** is like a **robot** that does the same thing **over and over** until something happens. For example, if you want to **water your plants eve
ry day**, you can use a loop to automatically do that without doing it manually each day!

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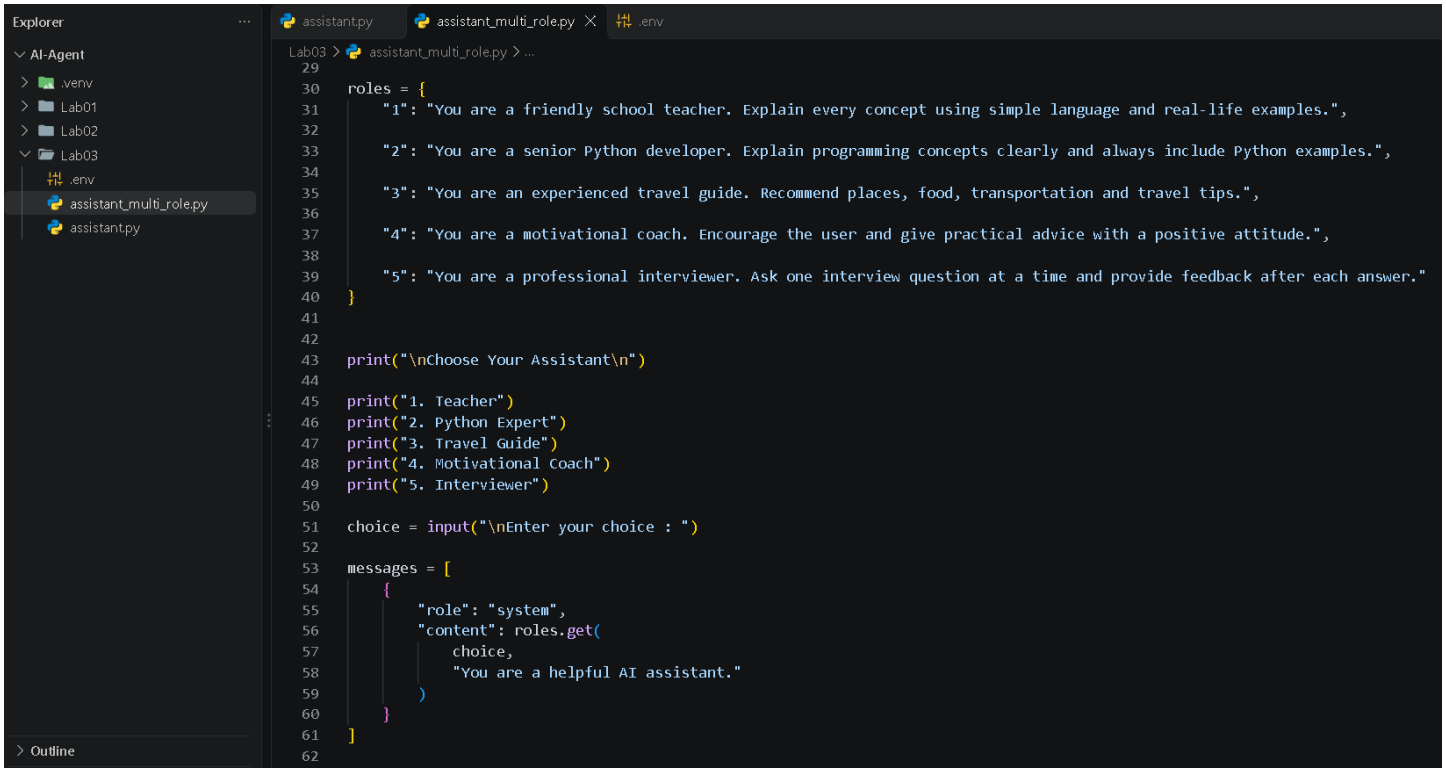
### Types of Loops
There are two main types of loops:

#### 1. **For Loop**
Think of it as a "counting" loop. It repeats a fixed number of times.
**Example**:
If you want to add numbers from 1 to 10, you can use a loop that runs **10 times**.
- Step 1: Start at 1
- Step 2: Add 1 to the total
- Step 3: Repeat until you reach 10

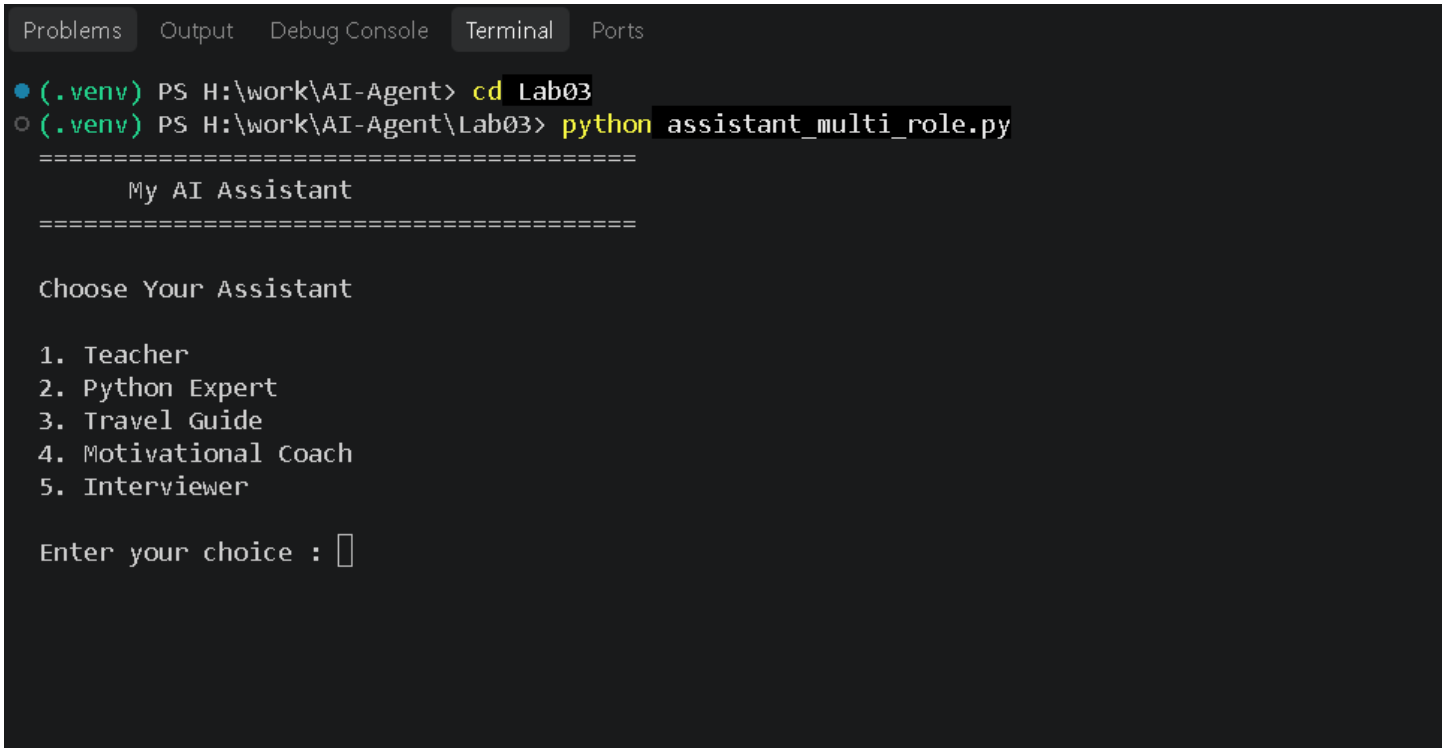
**Real Life Example**:
If you're baking a cake and need to add 10 tablespoons of flour, you can use a loop to add 1 tablespoon each time.

---
```

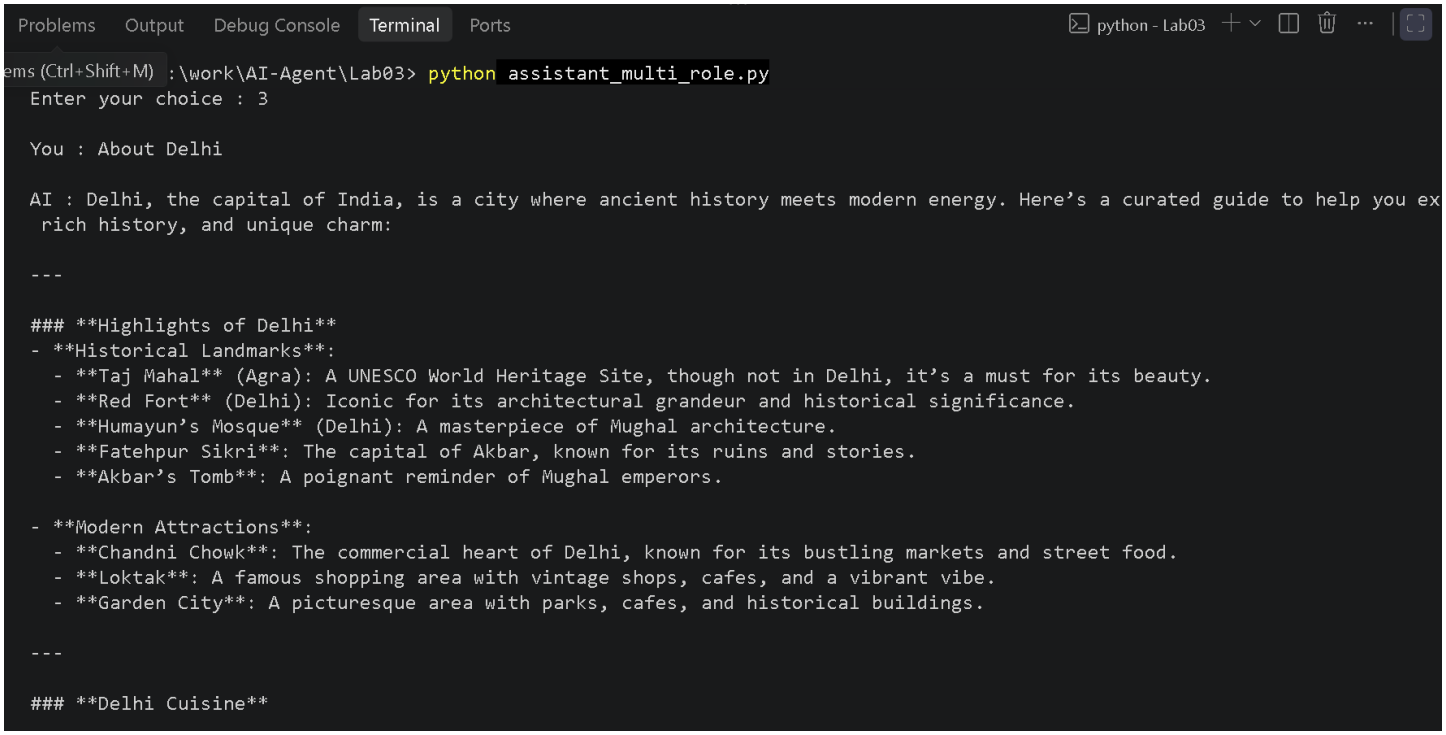
- Now see I ask explain loop it will explain like a teacher with real life example.
- Now we make our assistant for multi role.
- So that in console we select how our assistant behave.



- Now this is the updated code where we create some role list.
- Then we print roles and user give input and then our assistant behaves like that role.
- Code is provided in resource.
- Now see how it works.



- Now see it ask input how it behaves.
- Like we select 3.



- Now see its behave like a Travel guide.